

fox2db — controller model

Sofar / Waveshare ESP32-S3 6CH

Load-following PV-surplus controller for a solar plant with two battery stores. It switches a *quantized* electrical load (EBox) so that the grid exchange stays near zero (“sweet spot”), without disturbing the autonomous Sofar battery.

Classification

Discrete-time, nonlinear **hybrid automaton** (Mealy machine with hysteresis) — **not a PID, not an LTI system**. The actuator is discrete (8 power levels via 3 relay bits), hence dead-band + rate limiting instead of a continuous control signal. Cycle: 60 s.

Mathematical model

Power map and (power-sorted) rank:

$$P = (0, 3000, 3650, 6650, 3900, 7100, 7800, 11400) \text{ W}$$

$$\Pi_{\text{asc.}} = (0, 3000, 3650, 3900, 6650, 7100, 7800, 11400)$$

Step recursion, state $x_k = (s, c, E)$:

$$e_{\text{eff}} = \max(e_k, P_{s_{k-1}}) \quad (s_{k-1} > 0), \quad E_k = p_k + e_{\text{eff}} + b_k$$

$$\tau_k = \begin{cases} 0 & E_k < E_{\min} \\ Q(E_k + G) & \text{otherwise} \end{cases} \quad Q(B) = \max\{s : P_s \leq B\}$$

Up-ramp (max. one power step per cycle):

$$\tilde{\tau}_k = \min_{\text{rank}}(\tau_k, \hat{s}(\text{rank}(s_{k-1}) + 1))$$

Switching / protection logic (order = priority):

Rule	Condition	Action
EMERGENCY_FORCE	$\text{pcc} < -1020$ ($= -(G+120)$) and down	switch now
SWEET_SPOT_HOLD	up and $ \text{pcc} < 160$	hold
TREND_BLOCK	up and $\dot{E} < -20 \text{ W/s}$	hold
BAT_GUARD_BLOCK	up and $b_1 < -110 \text{ W}$	hold
STABILIZING	down and $c < 2$ cycles	hold
HYSTERESIS	down and $ \Delta P < 505 \text{ W}$	hold

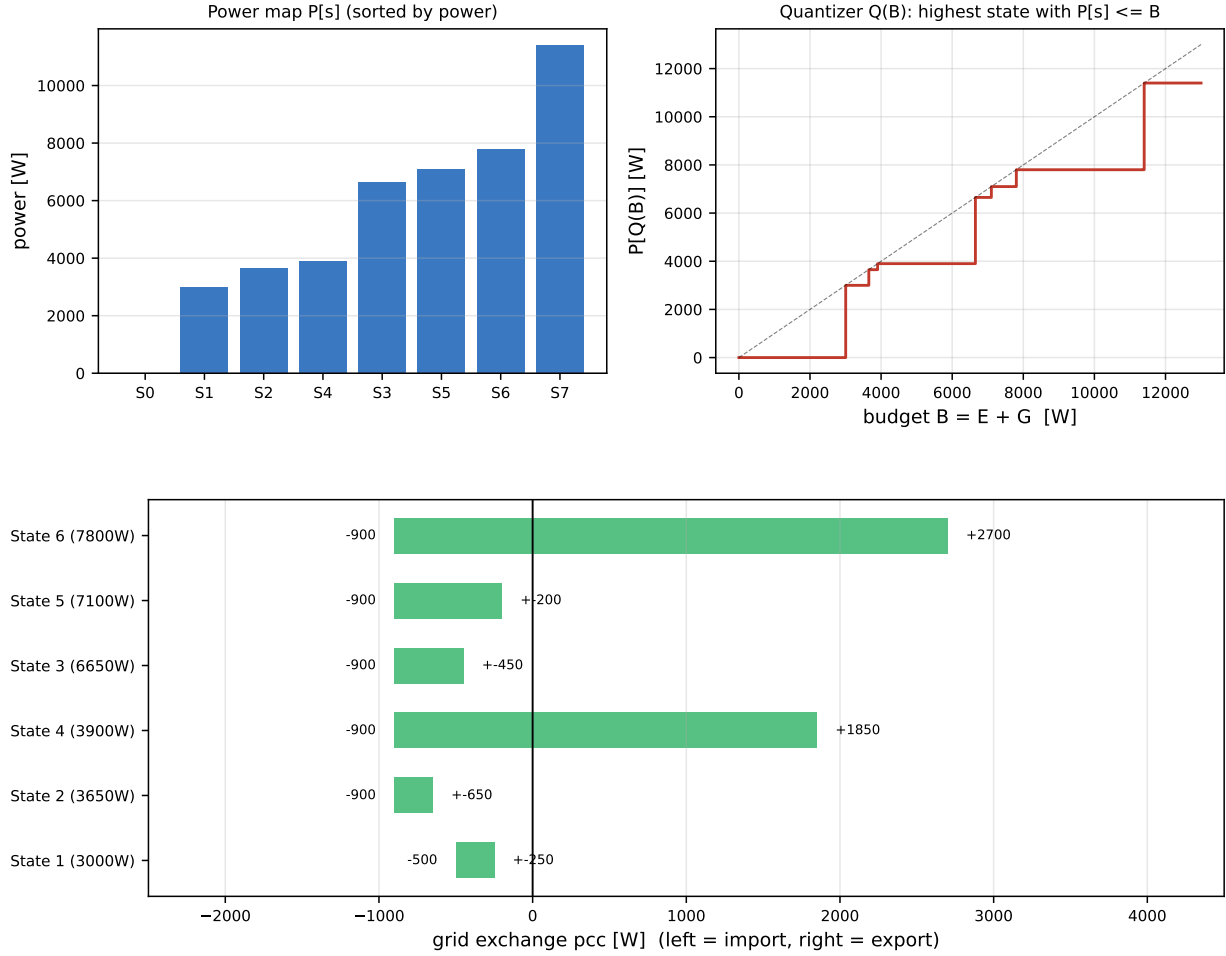
Constants: $E_{\min} = 2500$, $G = 900$, $H = 505$, $N = 2$, $\text{EMERG} = G + 120$ [W].

Dead-band (equilibrium, load on, $e_{\text{eff}} = P_s$):

$$\max(-G, E_{\min} - P_s) < p < (\Pi^+(s) - P_s) - G$$

Simulation

(a) Ramp-up: rate limiting allows only one power step per cycle. (b) Load step in the sweet spot: -2500 W from cycle 5 (heat pump) is shed within a few cycles.



Dead-band per state: lower edge = $\max(-G, E_{\min} - P_s)$, upper edge = next power level $-P_s - G$.

Plant limits

Fuse $3 \times 50 \text{ A @ } 230 \text{ V} = 34.5 \text{ kW}$; PV peak 35 kWp; Bat1 (Sofar) 5 kWh; Bat2 (EBox) 30 kWh. EBox max (state 7) 11.4 kW = 33 % of the fuse. Bat2 @ state 7: +0.63 %/min ($\rightarrow$ full in $\sim 2.6 \text{ h}$).

Validation

`step_full_literal` was checked one-step-ahead against the ESP's actual decisions (`pv_decision_log`): **55/55 state match (100 %)**, excess formula 54/55. Note: the C code compares by state *number* for the up-ramp rather than the power *rank*; since P is not monotone in the number (state4 = 3900 < state3 = 6650 W), this deviates in edge cases from the rank-monotone model. `step_full_literal` mirrors the C code 1:1.

